CIAM ID – MDM Linking Tables

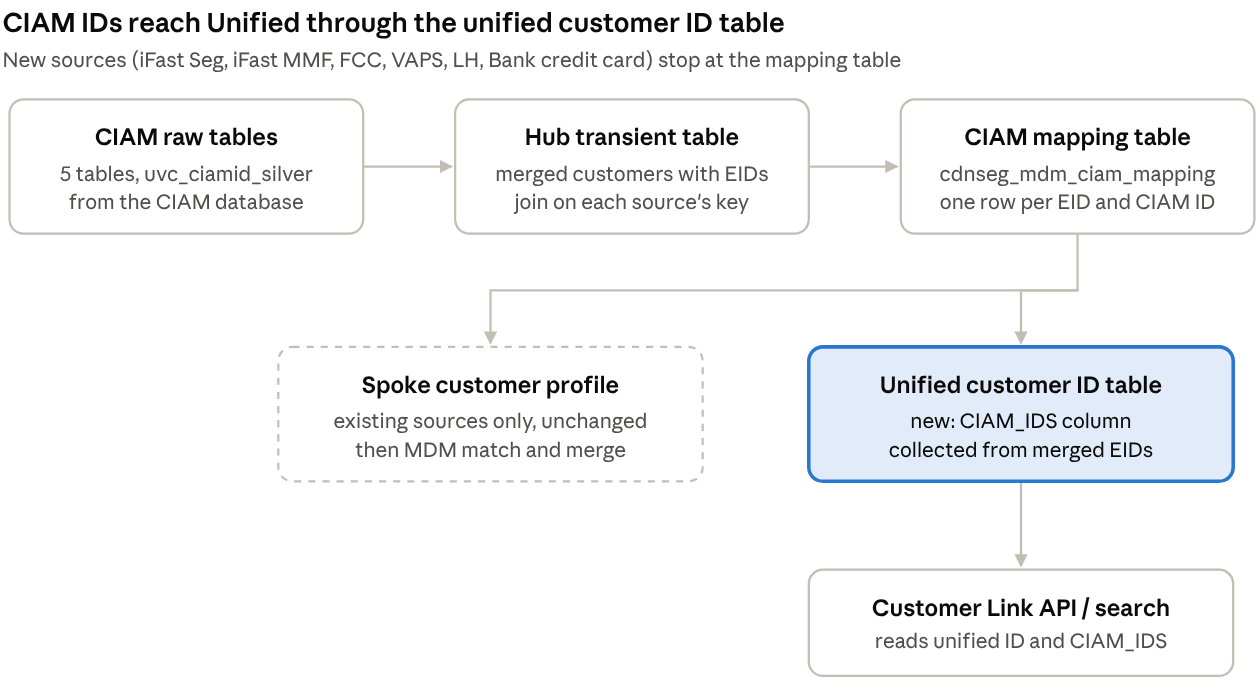
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To link a unified customer (ECID) to their CIAM IDs, we add a **CIAM_IDS** column to the unified customer ID table. It's filled from the CIAM mapping table by matching each of the customer's merged enterprise IDs. The CIAM mapping table itself doesn't change. CIAM status and provider (portal) stay in the CIAM raw tables. *Subpage of CIAM ID Mapping.*

1. How the three tables link

#	Table	Grain	Gives
1	`ops_prod_catalog.mdm_hub.cdnseg_mdm_ciam_mapping`	One row per CIAM ID and source customer	ENTERPRISE_ID → CIAMID
2	`ops_prod_catalog.mdm_hub.cdnseg_unified_customer_id`	One row per unified customer	UNIFIED_CUSTOMER_ID → list of MERGED_EID
3	CIAM raw tables in `ops_prod_catalog.uvc_ciamid_silver`	One row per CIAM link	Account number → CIAMID, with provider and status

Path: UNIFIED_CUSTOMER_ID → each MERGED_EID → ENTERPRISE_ID in the mapping table → CIAMID, stored back on the unified customer ID table as CIAM_IDS. A consumer that needs status or provider joins to table 3 on CIAMID.



CIAM ID linking flow · current and proposed

Existing sources keep their current path back to the spoke customer profile (dashed). New sources stop at the mapping table, and the new CIAM_IDS column on the unified customer ID table is what Unified reads.

2. CIAM mapping table

`ops_prod_catalog.mdm_hub.cdnseg_mdm_ciam_mapping` holds the record-level link between a source customer and its CIAM ID. No history is kept: rows are inserted or overwritten. New sources (iFast Seg, iFast MMF, FCC, VAPS, LH, Bank credit card) only update this table, joined at the hub's transient table once the enterprise ID exists; nothing is pushed back to the spokes.

```
DESCRIBE TABLE ops_prod_catalog.mdm_hub.cdnseg_mdm_ciam_mapping;
```

Column	Data type	Meaning
CIAMID	string	Customer's CIAM (online) ID
ENTERPRISE_ID	string	MDM enterprise ID of the source customer; null until MDM has the customer

Column	Data type	Meaning
SOURCE_CUSTOMER_NUM	string	Customer number in the source system (for AMAC it holds the enterprise ID)
SOURCE_SYSTEM_TP_CD	string	Source system code, for example 27 bank, 14 CLAS, 57 AMAC
SOURCE_SYSTEM_TP_VALUE	string	Source system name, for example BANK, CLAS
LAST_UPDATE_DT	timestamp	When the row was last written

Example rows (from the existing CIAMID-MDM Integration Flow page; replace with 2–3 real rows from a quick `SELECT`)

CIAMID	ENTERPRISE_ID	SOURCE_CUSTOMER_NUM	SOURCE_SYSTEM_TP_CD	SOURCE_SYSTEM_TP_VALUE	LAST_UPDATE_DT
123	234	ABC11	27	BANK	2024-10-04
123	456	ABC22	14	CLAS	2024-10-05
456		ABC33	14		

One CIAM ID (123) can link to customers in different BUs, and a row can have no enterprise ID yet (456).

3. Unified customer ID table: add CIAM_IDS

`ops_prod_catalog.mdm_hub.cdnseg_unified_customer_id` groups the enterprise IDs that belong to one person. The UNIFIED_CUSTOMER_ID is itself one of its MERGED_EID values.

Today

MDM_ID	UNIFIED_CUSTOMER_ID	MERGED_EID (first values)
MDM00000034G7Y	32696496431855980	13758124703592180, 25997594988017288, 32696496431855980, 33170339023813242, …
MDM00000035GU0	16916783932906830	16916783932906830, 19782999314925256, 30925367346987151, …

MDM_ID	UNIFIED_CUSTOMER_ID	MERGED_EID (first values)
MDM00000035L3Q	64630339314505166	24139683346730319, 64630339314505166, 8823428796227898

Proposed: new column

Column	Data type	Meaning
CIAM_IDS	array<string>	Every CIAM ID linked to any of the customer's merged enterprise IDs. A list, because one person can have several CIAM IDs (one per portal, or several logins)

```
ALTER TABLE ops_prod_catalog.mdm_hub.cdnseg_unified_customer_id
  ADD COLUMNS (CIAM_IDS ARRAY<STRING>);
```

Example after the change (CIAM IDs illustrative)

MDM_ID	UNIFIED_CUSTOMER_ID	MERGED_EID	CIAM_IDS
MDM00000034G7Y	32696496431855980	13758124703592180, 25997594988017288, …	e3b479b1-…, 99e84602-…
MDM00000035L3Q	64630339314505166	24139683346730319, 64630339314505166, 8823428796227898	9364660e-…

How it's populated (runs after the unified customer ID and CIAM mapping tables refresh)

```
MERGE INTO ops_prod_catalog.mdm_hub.cdnseg_unified_customer_id t
USING (
  SELECT u.UNIFIED_CUSTOMER_ID,
         array_sort(collect_set(m.CIAMID)) AS CIAM_IDS
  FROM (
    SELECT UNIFIED_CUSTOMER_ID, explode(MERGED_EID) AS ENTERPRISE_ID
    FROM ops_prod_catalog.mdm_hub.cdnseg_unified_customer_id
  ) u
  JOIN ops_prod_catalog.mdm_hub.cdnseg_mdm_ciam_mapping m
    ON m.ENTERPRISE_ID = u.ENTERPRISE_ID
  GROUP BY u.UNIFIED_CUSTOMER_ID
) s
ON t.UNIFIED_CUSTOMER_ID = s.UNIFIED_CUSTOMER_ID
WHEN MATCHED AND NOT (t.CIAM_IDS <=> s.CIAM_IDS)
  THEN UPDATE SET t.CIAM_IDS = s.CIAM_IDS;
```

Notes:

- The row is only rewritten when its list of CIAM IDs changes.
- A customer who loses all CIAM IDs isn't in the source query, so a separate step must clear their CIAM_IDS.
- If MERGED_EID is stored as a JSON string rather than an array, use `explode(from_json(MERGED_EID, 'array<string>'))`.
- If the unified customer ID table is rebuilt (overwritten) on each run, add CIAM_IDS in that build job instead of a separate merge.

4. CIAM raw data: provider and status

The raw tables show which provider (portal) a CIAM ID belongs to and whether the link is active. One CIAM ID can be linked to several customers, and a CIAM ID can be reassigned after a customer goes inactive, so consumers may need the status.

```
SELECT id_a.IDENTITY_ID AS ACCOUNT_NUMBER,
       id_b.IDENTITY_ID AS CIAMID,
       prv_a.NAME       AS PROVIDER_NAME,
       st.NAME          AS CIAM_STATUS
FROM ops_prod_catalog.uvc_ciamid_silver.XIDENTITYMAP mapping
LEFT JOIN ops_prod_catalog.uvc_ciamid_silver.CDPROVIDERRELSTTP st
  ON mapping.STATUS_TP_CD = st.PROVIDERREL_ST_TP_CD
LEFT JOIN ops_prod_catalog.uvc_ciamid_silver.XIDENTITYREF id_a
```

```
  ON mapping.IDENTITY_A_ID = id_a.XIDENTITY_REF_PK
LEFT JOIN ops_prod_catalog.uvc_ciamid_silver.XIDENTITYREF id_b
  ON mapping.IDENTITY_B_ID = id_b.XIDENTITY_REF_PK
LEFT JOIN ops_prod_catalog.uvc_ciamid_silver.CDPROVIDERSYSTP prv_a
  ON id_a.PROVIDER_SYS_TP_CD = prv_a.PROVIDER_SYS_TP_CD
LEFT JOIN ops_prod_catalog.uvc_ciamid_silver.CDPROVIDERSYSTP prv_b
  ON id_b.PROVIDER_SYS_TP_CD = prv_b.PROVIDER_SYS_TP_CD
LEFT JOIN ops_prod_catalog.uvc_ciamid_silver.CDSOURCESYSTP src_b
  ON prv_b.SOURCE_SYS_TP_CD = src_b.SOURCE_SYS_TP_CD
WHERE src_b.NAME = 'ciam'
  AND prv_a.NAME IN ('iic-css-segfund', 'iic-hd-secureserve')
  -- AND st.NAME = 'ACTIVE'   (no status filter today)
;
```

ACCOUNT_NUMBER	CIAMID	PROVIDER_NAME	CIAM_STATUS
99e84602-b359-4f7a-bb7b-ab322e5b63ce@manulife.com	99e84602-b359-4f7a-bb7b-ab322e5b63ce	iic-css-segfund	ACTIVE
177774-6822759294	e3b479b1-1beb-4813-8eb0-063645954640	iic-hd-secureserve	ACTIVE
1777MP-6200903400	9364660e-6810-46a1-be6e-7084afa89247	iic-hd-secureserve	ACTIVE

What the samples show:

- **iic-css-segfund** (seg fund) identities are an email-style login; the part before "@manulife.com" is the CIAM ID itself, so a different key is needed to reach the source customer.
- **iic-hd-secureserve** (LH) identities are account numbers such as 177774-6822759294, 1777MP-…, 1777FM-… and 1777FG-….
- CIAM IDs are GUIDs, which answers the earlier question about the hash-like format.

5. Open points for Theo

- **Column or join:** confirm the CIAM_IDS column on the unified customer ID table (this page) rather than keeping the tables separate and letting consumers join or use a view. Both were discussed.
- **Inactive CIAM IDs:** should CIAM_IDS include every linked CIAM ID, or only active ones? Today no status filter is applied.
- **Plain list or details:** is a list of CIAM IDs enough, or should each entry carry its source system (and status) as a struct?
- **Provider name:** is it needed anywhere, or can consumers join to the raw data when they need it?
- **Search and match:** does it return CIAM_IDS with the unified customer, or does the consumer look them up? If it returns them, who orchestrates that?